ML pipeline (2/2)

MADRE: Méthodologie pour l'Analyse de Données Réseaux
(Methodology for Network Data Analysis)

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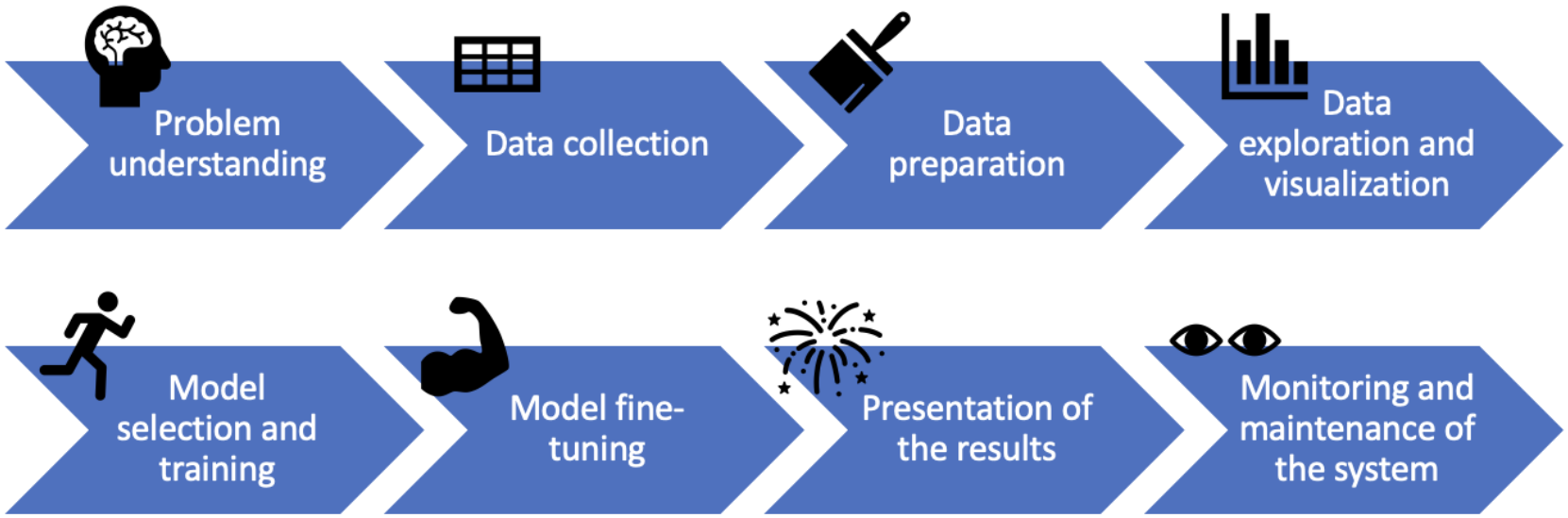
Outline

1. Summary
2. Model selection and training
3. Model fine-tuning
4. Presentation of the results
5. Monitoring and maintenace of the system
6. Let's play

High level overview



High level overview

Problem understanding:
The process of deeply understanding the real-world problem you are trying to solve with machine learning, before jumping into data collection, modeling, or algorithms.
Data collection:
It is the phase in which the raw data that your model will learn from are gathered.
Data preparation:
Data preprocessing, or data preparation, transforms the raw data from the previous step into clean data that is ready for analysis (Data cleaning, Data transformation, ...)
Data exploration and visualization:
It is the step where you get to know your data. It is about uncovering patterns, spotting problems, and forming hypotheses before you train any machine learning model.

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Model selection

Model selection means choosing the right algorithm (or set of algorithms) to solve your ML problem.
This step has to take into account:

- the type of problem,
- the data size and quality,
- the interpretability requirements,
- the computation constraints,
- the training time,
- ...

Model training

Model training means feeding your training data into the selected model so it can learn patterns.
The dataset is splited (during the data preparation phase) in:

- Training Set: $70 - 80\%$
- Validation Set: $10 - 15\%$
- Test Set: $10 - 15\%$

The training set is used to train the model, the validation set is used to evaluate during the training and the test set is used for final evaluation[1].

[1] Please refer to the lesson on cross-validation.

Example

A company runs a large-scale network and wants to detect network intrusions (cyberattacks) in real-time.

In problem understanding phase we concluded that:

- It is a Network Intrusion Detection (NID) problem so the objective is to detect if a network event (e.g., a packet, session, or connection) is normal or malicious in real-time.
- it is typically a classification problem (supervised or not depending on the collected data)

Let us consider a labeled dataset.

Example

A company runs a large-scale network and wants to detect network intrusions (cyberattacks) in real-time.

Model	Pros	Cons
Logistic Regression	Fast, interpretable	Poor with complex patterns
Decision Trees	Fast, interpretable	Prone to overfitting
Random Forest	High accuracy, handles non-linearities	Slower, more memory
SVM	Good for small to medium data	Slow on large-scale
Neural Networks	Can model complex patterns	Needs more data, harder to interpret

Example

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Train the Model:

- Train / Validation / Test or cross-validation technique
- Fit the model on the training data

Evaluate the Model: Use relevant metrics (not just accuracy). Let us call:
TP:True Positive, FN: False Negative, FP: False Positive, TN: True negative
Interesting metrics to consider are:

- Precision : High precision means few false alarms
- Recall: High recall means few attacks are missed
- False Positive Rate : Out of all normal traffic, how many were wrongly flagged as attacks
- ...

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Model fine-tuning

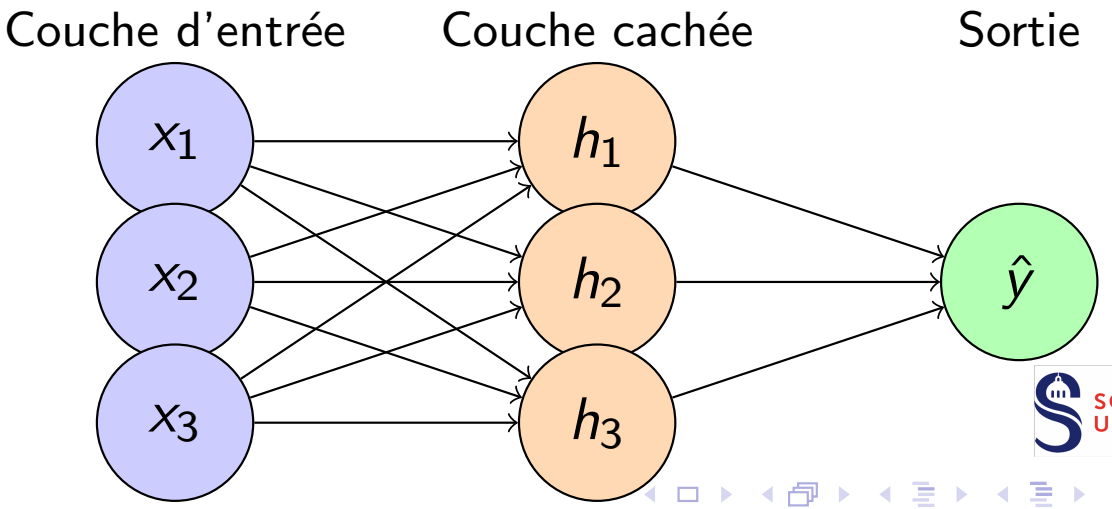
Model fine-tuning is the process of optimizing your trained model by adjusting its hyperparameters and, in some cases, its architecture to achieve better performance.
This is a critical step for improving your model's performance after it has been trained and validated.
Goals:

- Improve generalization (reduce overfitting or underfitting)
- Optimize performance metrics
- Reduce training time or resource consumption

Hyperparameters you might tune: an example

For Neural Networks (NNs):

- Architecture: number of layers, neurons per layer
- Training: learning rate, batch size, optimizer
- Epochs: how many full passes through the data
- ....



Methods of Fine-tuning

- Grid Search
  Exhaustively tests all combinations of provided hyperparameters. Great for small search spaces.
- Random Search
  Randomly selects combinations to try. More efficient for large search spaces.
- Bayesian Optimization / Automated Tuning
  Uses past evaluations to choose the next best set of hyperparameters to try.
- Manual Tuning
  Based on experience and intuition. Good for small projects or when you have domain knowledge.

Example

A company runs a large-scale network and wants to detect network intrusions (cyberattacks) in real-time.

We provide a Random Forest classifier.

Hyperparameters:

- Number of trees (`n_estimators`)
- Maximum tree depth (`max_depth`)
- Minimum samples per leaf (`min_samples_leaf`)

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Presentation of the results

Clearly communicate your model's findings, performance, and insights to technical and non-technical stakeholders, so they can make informed decisions.

A detailed presentation can include:

- Problem Statement
- Data Summary
- Model(s) Tried
- Evaluation Metrics
- Visualizations
- Business/Real-world Interpretation
- Model Limitations
- Next Steps

Presentation of the results

Problem Statement:
Briefly restate the goal of the project: What problem are we solving? Why is it important?

Data Summary:
Explain the data you used:

- Source of the data
- Number of samples, features
- Class balance (normal vs attack)
- Preprocessing steps (scaling, encoding, balancing)

Model(s) Tried:
List the models you experimented with, why you chose them, ...

Presentation of the results

Evaluation Metrics:
Present the key metrics
Eventually use tables and charts to compare models.
Visualizations:
Provide plots to present the results.
Pay attention to:

- Use clear labels and titles
- Use colors carefully
- Tailor visuals for your audience:
  - Executives: Use summaries and bar charts
  - Data scientists: Show curves, matrices, and tuning results

Presentation of the results

Business/Real-world Interpretation:
Translate the metrics into real-world meaning.
Model Limitations:
Be honest:

- Any overfitting?
- Data issues ?
- Real-time latency concerns?
- Interpretability concerns?
- ...

Next Steps:
Explain if next step are planned, for example improving data quality, trying new (more advanced) models, ...

Example

A company runs a large-scale network and wants to detect network intrusions (cyberattacks) in real-time.

Problem statement

"The objective is to detect if a network event (e.g., a packet, session, or connection) is normal or malicious in real-time. The current manual monitoring is too slow and prone to error."

Data Summary

"The dataset includes 10 million network flow records collected over a 2-week period. 95% of the flows are labeled as normal, and 5% are intrusions (mainly DDoS and port scanning attempts)."

Example

A company runs a large-scale network and wants to detect network intrusions (cyberattacks) in real-time.

Model(s) Tried

"We used a random forest model."

Evaluation Metrics

"The random forest model achieved a recall of 98% and a precision of 92%."

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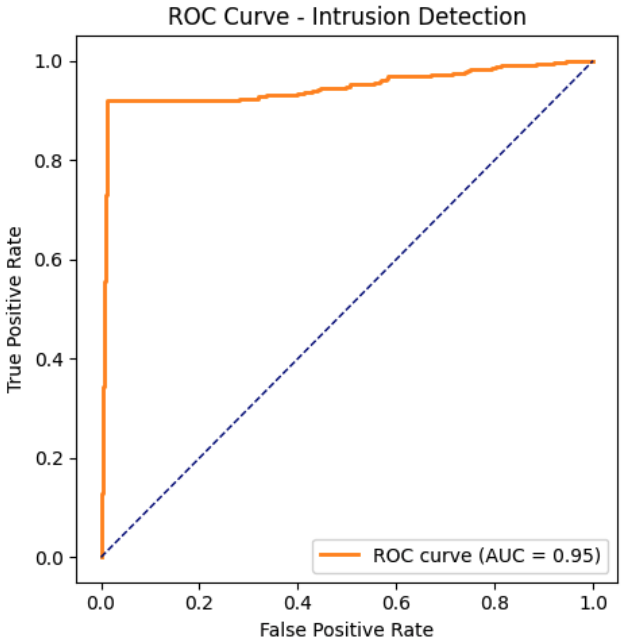
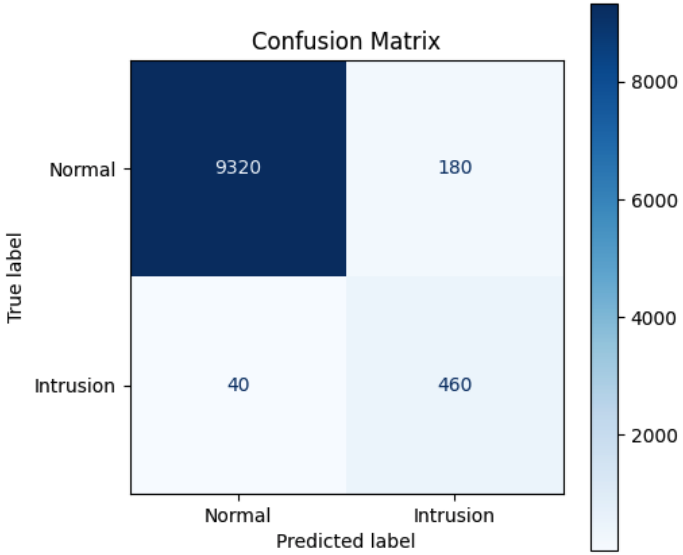
Visualisations





Example

A company runs a large-scale network and wants to detect network intrusions (cyberattacks) in real-time.

Business interpretation

Translate the technical results into business-relevant insights for executives, managers, and non-technical stakeholders.
You can include: how the model improves existing intrusion detection capabilities, cost/time saved due to reduced analyst workload, faster incident response, ...
"By reducing false positives by 40%, the model can save the security operations center team an estimated 20 hours per week in manual triage."

Example

A company runs a large-scale network and wants to detect network intrusions (cyberattacks) in real-time.

Model limitations

"The model struggles to detect DoS and DDoS intrusions due to their similarity to normal traffic."

Next steps

"Next, we plan to integrate this model into the Security Information and Event Management pipeline and begin online retraining using daily logs."

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Monitoring and maintenace of the system

The monitoring and maintenance step involves tracking, evaluating, and updating a machine learning model after it has been deployed to ensure it remains accurate, relevant, and reliable. It is important because:

- Data changes over time
- Behavior changes, users and environments evolve
- Bad predictions can damage business or user trust
- Some systems must meet ongoing legal/ethical standards
- ...

Monitoring of the system

Monitor Input Data (Data Drift)
Check if incoming data has different patterns from training data. Check: Feature distributions (mean, std, value ranges), Missing values or schema mismatches, Categorical value changes, ...

Monitor Model Performance (Concept Drift)
Check the metrics you used to evaluate the model, changes in output probability distributions, performance on recent labeled data, ...

Monitor System Health

- API latency, success/failure rates
- Memory, CPU, throughput
- Prediction volume and frequency

Alerts and Dashboards

- Set thresholds on key metrics
- Trigger alerts to teams
- Use dashboards to visualize real-time performance

Maintenance of the system (1/2)

Once you detect an issue, you act. This is the maintenance part.

Retraining the Model

Use new data to retrain the model, Update periodically, Revalidate performance before redeployment

Version Control

Track versions of: Model code, Data used for training, Hyperparameters, Model performance, ...

Human Feedback Loop

Allow users or analysts to give feedback and use them to improve future training

Maintenance of the system (2/2)

Deployment Management

- Use A/B testing or canary deployments when releasing new models:
  A/B testing is used to compare the performance of two (A/B) or more models in a real-world setting, not just on test data.

  Canary deployment in ML means slowly rolling out a new model version to a small subset of users or systems before full deployment.
- Rollback easily if performance drops

  Rolling back an ML model means reverting to a previous version if the new model causes issues in production.
- Keep multiple versions live if needed (e.g., per region or user group)

Example

A company runs a large-scale network and wants to detect network intrusions (cyberattacks) in real-time.

Moinitoring of the system
You detect that the false positive rate increases from 1.8% to 5% over a week. This might indicate model drift or new types of traffic not seen during training.
Maintenance of the system
Keep your model up-to-date with current network patterns and attack signatures.

- Scheduled retraining: Weekly or monthly using newly labeled data.
- Trigger-based retraining: When a drift or performance drop is detected.

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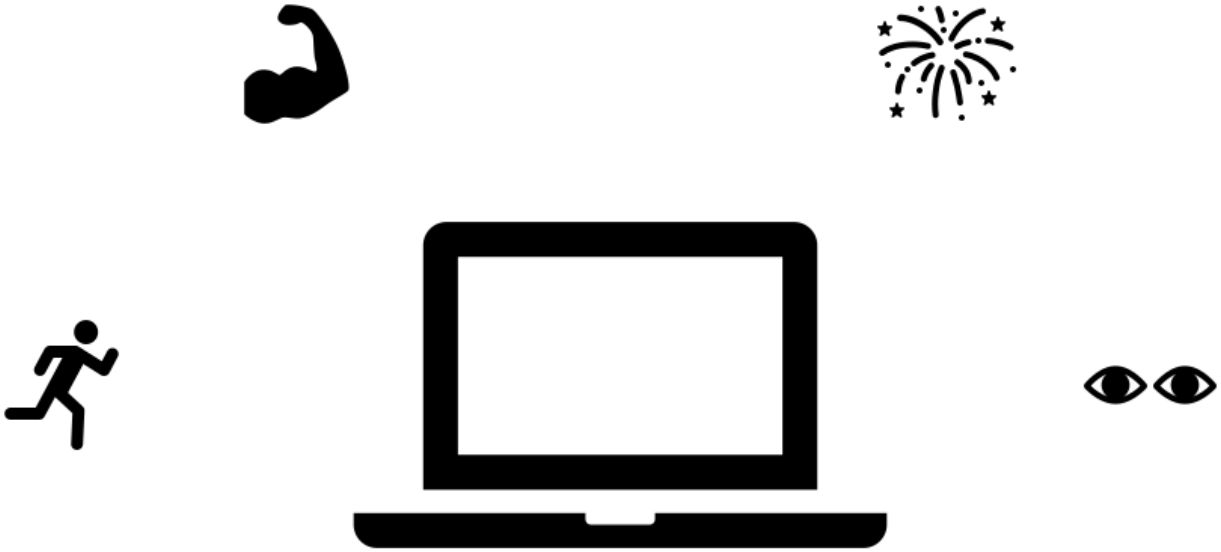
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